

SUPPLEMENT TO:
THE HITCHHIKER’S GUIDE TO MARKUP ESTIMATION: ASSESSING ESTIMATES
FROM FINANCIAL DATA

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APPENDIX A: THEORY APPENDIX

In this appendix, we first outline how the output elasticity of V_{it} is estimated. We start from the ideal case in which a researcher observes prices, such that output can be measured by quantity (A.1). The main text discusses the case where price is not observed. We then discussed the small sample properties of the estimator (A.2) and the case of measurement errors (A.3). We also show that the main results extend to more general frameworks (A.4) with a translog production function (A.4.1), several inputs (A.4.2), with persistent productivity (A.4.3), and with all of this together (A.4.4). We generalize our results on markup estimation when revenue is used in place of quantity for translog production functions (A.5).

A.1. *Identification with Price and Quantity Data*

We here cover the estimation of α if revenue, prices, and quantities are observable. Our estimator for α builds on the two-stage GMM estimator of [Akerberg et al. \(2015\)](#) to accommodate imperfect competition. The first stage purges the quantity of equation (3) of the measurement error and unobserved productivity shocks η_{it} . The second stage estimates the output elasticity α using an instrumental-variable generalized method of moments (IV-GMM). We focus in this section on the second stage – as it performs the actual production function identification. We introduce measurement errors and a first stage in Section A.3.

Absent measurement error, the production function is $y_{it} = \alpha v_{it} + \omega_{it}$. Least-square regressions of v_{it} on output y_{it} will be biased, as unobserved productivity ω_{it} (the residual in the regression) affects firms’ choice of v_{it} . Following the literature, we can construct an estimator to identify α by instrumenting v_{it} by v_{it-1} :

DEFINITION 1: *The instrumental var. GMM (IV-GMM) estimator $\hat{\alpha} \in \mathbb{R}$ is such that the moment $\mathbb{E}[\hat{\omega}_{it} v_{it-1}]$ is equal to 0 where $\hat{\omega}_{it} = y_{it} - \hat{\alpha} v_{it} = (\alpha - \hat{\alpha}) v_{it} + \omega_{it}$.*

It is straightforward to solve for $\hat{\alpha}$ in closed form by substituting $\hat{\omega}_{it}$ into the moment condition:¹ $(\alpha - \hat{\alpha}) \mathbb{E}[v_{it} v_{it-1}] = 0$, which uses the fact that that productivity ω_{it} is orthogonal to

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¹In the above definition, the expectation operator $\mathbb{E}$ denotes the limit of the empirical average across observations. We therefore study the asymptotic properties of the GMM estimator, which allows us to keep the argument as tractable as possible.

v_{it-1} , such that $\mathbb{E}[\omega_{it}v_{it-1}] = 0$. It follows that as long as v_{it-1} is a relevant instrument for v_{it} (that is, $\mathbb{E}[v_{it}v_{it-1}]$ differs from zero), the only solution is that $\hat{\alpha} = \alpha$. Our estimator $\hat{\alpha}$ thus converges to the true elasticity α .

What ensures that the lagged variable input is a relevant instrument? As we have assumed – for now – that productivity is not persistent, autocorrelation in v_{it} comes from other sources.² The cost-minimizing firm’s first-order condition for v_{it} summarizes the candidate drivers: $v_{it} = (1 - \alpha)^{-1}(\omega_{it} + mc_{it} - w_t)$.

It follows that persistence in v_{it} has to either come from persistence in the input price w_t or from log marginal costs mc_{it} . Marginal costs equal P_{it}/μ_{it} , both of which are determined in equilibrium by the demand system and the strategic interactions among firms. Hence, any persistence in output price or markups will contribute to persistence in the variable input and thus to identification of the production function. Persistence in input prices w_t is a source of persistence in variable inputs regardless of the mode of competition, providing a further source of identification of α (a point previously made by, e.g., [Gandhi et al. 2020](#)). We conclude that the parameters of the production function in our simple framework are identified under imperfect competition as long as there is persistent variation in markups, output prices or input prices.³

A.2. Finite Sample Estimator and its Asymptotic Variance

In this section we derive the estimator for a finite sample. We also use this derivation to compute the asymptotic variance of the GMM estimator.

DEFINITION: *The GMM estimator is $\hat{\alpha}$ such that $\sum_{i,t} \hat{\omega}_{it}v_{it-1} = 0$ with $\hat{\omega}_{it} = y_{it} - \hat{\alpha}v_{it} = (\alpha - \hat{\alpha})v_{it} + \omega_{it}$.*

To solve for the estimator, we need to find the value of $\hat{\alpha}$ such that $\sum_{i,t} \hat{\omega}_{it}v_{it-1} = (\alpha - \hat{\alpha})\sum_{i,t} v_{it}v_{it-1} + \sum_{i,t} \omega_{it}v_{it-1} = 0$. As long as $\sum_{i,t} v_{it}v_{it-1} \neq 0$, the unique $\hat{\alpha}$ that solves this equation is $\hat{\alpha} = \alpha + \frac{\sum_{i,t} \omega_{it}v_{it-1}}{\sum_{i,t} v_{it}v_{it-1}}$, whose limit is α when the sample size increases, given that $\mathbb{E}[\omega_{it}v_{it-1}] = 0$.

Finally, let us derive the asymptotic variance of the GMM estimator. Using the (finite sample) expression of the estimator, we have $\sqrt{n}(\hat{\alpha} - \alpha) = \frac{\sqrt{n}\frac{1}{n}\sum_{i,t} \omega_{it}v_{it-1}}{\frac{1}{n}\sum_{i,t} v_{it}v_{it-1}}$. By the (weak) law of large numbers, $\frac{1}{n}\sum_{i,t} v_{it}v_{it-1} \xrightarrow{p} \mathbb{E}[v_{it}v_{it-1}]$, and, by the central limit theorem, $\sqrt{n}\frac{1}{n}\sum_{i,t} \omega_{it}v_{it-1} \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\omega_{it}^2v_{it-1}^2])$. The Slutsky theorem implies $\sqrt{n}(\hat{\alpha} - \alpha) \xrightarrow{d} \mathcal{N}\left(0, \frac{\mathbb{E}[\omega_{it}^2v_{it-1}^2]}{\mathbb{E}[v_{it}v_{it-1}]^2}\right)$; that is, $\text{Var}[\hat{\alpha}] \sim \frac{\mathbb{E}[\omega_{it}^2]\mathbb{E}[v_{it-1}^2]}{\sqrt{n}\mathbb{E}[v_{it}v_{it-1}]^2}$.

A.3. Adding Measurement Errors

As in the baseline framework, assume that firms produce y_{it} using the single variable input v_{it} while being subject to idiosyncratic productivity shocks ω_{it} . Furthermore, assume that the

²When we generalize our setup in Appendix A.4.3 (where productivity is persistent, e.g. a linear AR(1) process with coeff. ρ), we show that the necessary condition for identification is to have autocorrelation in $\tilde{v}_{it} = v_{it} - \rho v_{it-1}$. Persistence in productivity itself therefore does not aid identification.

³Note that this means that it is more straightforward to estimate the production function under imperfect competition than under perfect competition. Under perfect competition (where marginal costs equal prices), persistence in the variable input cannot come from the markup. If output prices are (e.g.) i.i.d., this means that the only source of persistence is the input price ([Gandhi et al. 2020](#)).

firms' output is observed subject to measurement error, or equivalently, that unexpected productivity shocks occur after input v_{it} is set. The measurement error is log-additive and denoted by η_{it} . All firms produce along $\tilde{y}_{it} = \alpha v_{it} + \omega_{it} + \eta_{it}$, where $\tilde{y}_{it}$ denotes observed output. We assume that measurement errors at time t are independent of the past value of the variable input; that is, $\mathbb{E}[\eta_{it}v_{it-1}] = 0$. If the econometrician ignores the presence of these measurement errors, the IV-GMM estimator is defined as follows:

DEFINITION 2: *The IV-GMM estimator is $\hat{\alpha} \in \mathbb{R}$ so that moment $\mathbb{E}[(\widehat{\omega_{it} + \eta_{it}})v_{it-1}]$ is equal to zero, where $\widehat{\omega_{it} + \eta_{it}} = \tilde{y}_{it} - \hat{\alpha}v_{it} = (\alpha - \hat{\alpha})v_{it} + \omega_{it} + \eta_{it}$.*

The GMM estimator is characterized by: $\mathbb{E}[(\widehat{\omega_{it} + \eta_{it}})v_{it-1}] = (\alpha - \hat{\alpha})\mathbb{E}[v_{it}v_{it-1}] = 0$, where we use the fact that $\mathbb{E}[\widehat{\omega_{it}}v_{it-1}] = 0$. The GMM estimator $\hat{\alpha}$ of the variable input's output elasticity is equal to α as long as $\mathbb{E}[v_{it}v_{it-1}] \neq 0$. The estimator is still consistent and measurement error only increases the variance of the composite error term $\omega_{it} + \eta_{it}$ in the production function. This point is known and has been discussed, for example, in [Blundell and Bond \(2000\)](#).

There are three advantages to purging the observed quantity from measurement errors. The first is that the increase in the variance of the composite error term $\omega_{it} + \eta_{it}$ in the production function raises the standard errors of the production function estimation. Indeed, a similar derivation to the one in [Appendix A.2](#) yields that the asymptotic variance of the estimator is $\text{Var}[\hat{\alpha}] \sim \frac{\mathbb{E}[v_{it-1}^2]}{n\mathbb{E}[v_{it}v_{it-1}]^2} (\mathbb{E}[\omega_{it}^2] + \mathbb{E}[\eta_{it}^2])$, which increases in measurement error variance.

The second advantage is that purging allows the econometrician to identify true productivity ω_{it} , which is relevant in many applications.

Third, measurement error can also impede the consistency of the IV-GMM estimator if ω_{it} is persistent with non-linear autoregressive terms ([Bond et al. 2021](#)). With persistent productivity, the moment conditions of the IV-GMM estimator have to be slightly altered to consistently estimate α (see [Appendix A.4.3](#)). For a linear AR(1) process of ω_{it} , the moment conditions are that lagged inputs v_{it-1} and estimated productivity $\widehat{\omega}_{it}$ are orthogonal to the innovation of the AR(1) process. For non-linear processes (e.g. quadratic, cubic), in the absence of measurement error, the additional moment conditions are that the higher-degree terms (e.g. $\widehat{\omega}_{it}^2$, $\widehat{\omega}_{it}^3$) are orthogonal to the AR(1) innovation. Measurement error, however, contaminates the productivity estimates $\widehat{\omega}_{it}$. This means that moment conditions with, e.g., $\widehat{\omega}_{it}^2$, $\widehat{\omega}_{it}^3$ contain higher-order moments of the measurement error. This prevents the moment conditions from holding at the true value of the output elasticity. For example, a common empirical assumption is that the productivity process is well-approximated by $\omega_{it} = \rho_1\omega_{it-1} + \rho_2\omega_{it-1}^2 + \xi_{it}$, where ξ_{it} are white-noise productivity shocks. In the presence of measurement error, the moment conditions $\mathbb{E}[v_{it-1}\widehat{\xi}_{it}] = 0$, $\mathbb{E}[\widehat{\omega}_{it-1}\widehat{\xi}_{it}] = 0$, and $\mathbb{E}[\widehat{\omega}_{it-1}^2\widehat{\xi}_{it}] = 0$, where $\widehat{\omega}_{it}$ is defined as before while $\widehat{\xi}_{it} \equiv \widehat{\omega}_{it} - \widehat{\rho}_1\widehat{\omega}_{it-1} - \widehat{\rho}_2\widehat{\omega}_{it-1}^2$, will not suffice. The problem is the non-linear moment condition $\mathbb{E}[\widehat{\omega}_{it-1}^2\widehat{\xi}_{it}] = 0$. To see this, consider the value of the moment at $\hat{\alpha} = \alpha$: $\mathbb{E}[\widehat{\omega}_{it-1}^2\widehat{\xi}_{it}] = \mathbb{E}[(\omega_{it} + \eta_{it} + (\alpha - \hat{\alpha})v_{it})^2\widehat{\xi}_{it}] = \mathbb{E}[\eta_{it}^2\widehat{\xi}_{it}] \neq 0$. It follows that the IV-GMM estimator does not estimate the production function parameters unless productivity follows a linear (dynamic) process.

A.4. Extensions

We now show that the identification results of our estimator are robust to several extensions that are common in practical applications. We study the case of the translog production func-

tion, the case of several inputs, the case of AR(1) productivity, and the case with all of these extensions together.

A.4.1. Translog Production Function

We first ease the assumption that output is log-linear by replacing the Cobb-Douglas production function with a translog specification: $y_{it} = \alpha v_{it} + \beta v_{it}^2 + \omega_{it}$. The other assumptions are unchanged. We assume that quantity is observed in this section. We leave the discussion of unobserved quantity and its implications for markup estimation to the dedicated Appendix A.5. Our aim is to identify α and β , in order to calculate size-dependent output elasticities of the variable input for the calculation of true markups $\mu_{it} = (\alpha + 2\beta v_{it})(P_{it}Y_{it})/(W_t V_{it})$. The least-squares estimation of the production function suffers from the same bias as before, which we address by instrumenting v_{it} and v_{it}^2 by their respective lags. Econometrically, estimating the translog production involves estimating a multivariate GMM regression with instrumental variables. Formally, we define the estimator as:

DEFINITION 3: *The GMM estimator is a pair $(\hat{\alpha}, \hat{\beta})$ such that $\mathbb{E}[\hat{\omega}_{it}v_{it-1}] = 0$ and $\mathbb{E}[\hat{\omega}_{it}v_{it-1}^2] = 0$ where $\hat{\omega}_{it} = y_{it} - \hat{\alpha}v_{it} - \hat{\beta}v_{it}^2 = (\alpha - \hat{\alpha})v_{it} + (\beta - \hat{\beta})v_{it}^2 + \omega_{it}$.*

It remains simple to solve for the estimator $(\hat{\alpha}, \hat{\beta})$ in our parsimonious setting. This involves solving the system of equations implied by the moment conditions:

$$\begin{aligned} \mathbb{E}[\hat{\omega}_{it}v_{it-1}] &= 0 \\ \mathbb{E}[\hat{\omega}_{it}v_{it-1}^2] &= 0 \end{aligned} \iff \begin{aligned} (\alpha - \hat{\alpha})\mathbb{E}[v_{it}v_{it-1}] + (\beta - \hat{\beta})\mathbb{E}[v_{it}^2v_{it-1}] &= 0 \\ (\alpha - \hat{\alpha})\mathbb{E}[v_{it}v_{it-1}^2] + (\beta - \hat{\beta})\mathbb{E}[v_{it}^2v_{it-1}^2] &= 0 \end{aligned}$$

This system can be rewritten in matrix form with $V(B - \hat{B}) = 0$, where

$$B - \hat{B} = \begin{pmatrix} \alpha - \hat{\alpha} \\ \beta - \hat{\beta} \end{pmatrix} \quad \text{and} \quad V = \begin{pmatrix} \mathbb{E}[v_{it}v_{it-1}] & \mathbb{E}[v_{it}^2v_{it-1}] \\ \mathbb{E}[v_{it}v_{it-1}^2] & \mathbb{E}[v_{it}^2v_{it-1}^2] \end{pmatrix}.$$

If the determinant of V is not zero, the GMM estimator on translog is identified and consistent, i.e. $\hat{\alpha} = \alpha$ and $\hat{\beta} = \beta$.

A.4.2. Several Inputs

In this extension, we assume that firms produce with two inputs: a variable input v_{it} and another input k_{it} . We assume that k_{it} is, in the terminology of the production function literature, dynamic. This means that firms face adjustment costs and other inter-temporal constraints when setting k_{it} , which leads firms to choose k_{it} before observing current productivity, i.e. $\mathbb{E}[\omega_{it}k_{it}] = 0$. The production function in logs reads $y_{it} = \alpha v_{it} + \beta k_{it} + \omega_{it}$. We are interested in estimating (α, β) . As k_{it} is set before productivity is observed, we only need to instrument the variable input with its lag. The estimation is therefore a GMM regression with one endogenous and one exogenous variable. For quantity, the estimator is

DEFINITION 4: *The GMM estimator is a pair $(\hat{\alpha}, \hat{\beta})$ so that $\mathbb{E}[\hat{\omega}_{it}v_{it-1}] = 0$ and $\mathbb{E}[\hat{\omega}_{it}k_{it}] = 0$, where $\hat{\omega}_{it} = y_{it} - \hat{\alpha}v_{it} - \hat{\beta}k_{it} = (\alpha - \hat{\alpha})v_{it} + (\beta - \hat{\beta})k_{it} + \omega_{it}$.*

Solving for the estimator $(\hat{\alpha}, \hat{\beta})$ implies solving for the following system of equations, defined by the moment conditions:

$$\begin{aligned} \mathbb{E}[\widehat{\omega}_{it}v_{it-1}] = 0 \\ \mathbb{E}[\widehat{\omega}_{it}k_{it}] = 0 \end{aligned} \iff \begin{aligned} (\alpha - \widehat{\alpha})\mathbb{E}[v_{it}v_{it-1}] + (\beta - \widehat{\beta})\mathbb{E}[k_{it}v_{it-1}] &= 0 \\ (\alpha - \widehat{\alpha})\mathbb{E}[v_{it}k_{it-1}] + (\beta - \widehat{\beta})\mathbb{E}[k_{it}^2] &= 0 \end{aligned}.$$

This system can be rewritten in matrix form, with $V(B - \widehat{B}) = 0$, where

$$B - \widehat{B} = \begin{pmatrix} \alpha - \widehat{\alpha} \\ \beta - \widehat{\beta} \end{pmatrix} \quad \text{and} \quad V = \begin{pmatrix} \mathbb{E}[v_{it}v_{it-1}] & \mathbb{E}[k_{it}v_{it-1}] \\ \mathbb{E}[v_{it}k_{it-1}] & \mathbb{E}[k_{it}^2] \end{pmatrix}.$$

As long as the determinant of V is not zero, the GMM estimator is identified and asymptotically consistent such that $\widehat{\alpha} = \alpha$ and $\widehat{\beta} = \beta$.

Using Revenue Instead of Quantity. When revenue, denoted r_{it} in log, is used as a proxy for quantity, the estimator can be defined as follows:

DEFINITION 5: *The GMM estimator is a pair $(\widehat{\alpha}, \widehat{\beta})$ so that $\mathbb{E}[\widehat{\text{tfpr}}_{it}v_{it-1}] = 0$ and $\mathbb{E}[\widehat{\text{tfpr}}_{it}k_{it}] = 0$, where $\widehat{\text{tfpr}}_{it} = r_{it} - \widehat{\alpha}v_{it} - \widehat{\beta}k_{it} = (\alpha - \widehat{\alpha})v_{it} + (\beta - \widehat{\beta})k_{it} + p_{it} + \omega_{it}$.*

The estimator is the solution of the following system of equations:

$$\begin{aligned} (\alpha - \widehat{\alpha})\mathbb{E}[v_{it}v_{it-1}] + (\beta - \widehat{\beta})\mathbb{E}[k_{it}v_{it-1}] + \mathbb{E}[p_{it}v_{it-1}] &= 0 \\ (\alpha - \widehat{\alpha})\mathbb{E}[v_{it}k_{it-1}] + (\beta - \widehat{\beta})\mathbb{E}[k_{it}^2] + \mathbb{E}[p_{it}k_{it}] &= 0, \end{aligned}$$

which admits a unique solution if the determinant of V is not zero. If the latter is satisfied, this unique solution estimator is $\widehat{B} = B + V^{-1}P$, where we denote the vector $P = (\mathbb{E}[p_{it}v_{it-1}], \mathbb{E}[p_{it}k_{it}])'$. As in the simple framework, the bias comes from the correlation of price with the instruments. This estimator will be asymptotically non-consistent if either $\mathbb{E}[p_{it}v_{it-1}]$ or $\mathbb{E}[p_{it}k_{it}]$ are different from zero.

A.4.3. Persistent Productivity

In this extension, we assume that total factor productivity follows a first-order autoregressive (AR1) process in logs. We assume that quantity is observed. We leave the discussion on the case when revenue is used in place of quantity to the general proof in Appendix A.4.4. The production function is still $y_{it} = \alpha v_{it} + \omega_{it}$, while the productivity process is $\omega_{it} = \rho\omega_{it-1} + \xi_{it}$. Below we define the GMM estimator $(\widehat{\alpha}, \widehat{\rho})$ using v_{it-1} and $\widehat{\omega}_{it-1}$ as an instrument for v_{it} and $\widehat{\omega}_{it}$.

DEFINITION 6: *The GMM estimator is a pair $(\widehat{\alpha}, \widehat{\rho})$ so that $\mathbb{E}[\widehat{\xi}_{it}v_{it-1}] = 0$ and $\mathbb{E}[\widehat{\xi}_{it}\widehat{\omega}_{it-1}] = 0$, where $\widehat{\omega}_{it} = y_{it} - \widehat{\alpha}v_{it} = (\alpha - \widehat{\alpha})v_{it} + \omega_{it}$ and $\widehat{\xi}_{it} = \widehat{\omega}_{it} - \widehat{\rho}\widehat{\omega}_{it-1} = \xi_{it} + (\alpha - \widehat{\alpha})(v_{it} - \rho v_{it-1}) + (\rho - \widehat{\rho})\omega_{it-1} + (\rho - \widehat{\rho})(\alpha - \widehat{\alpha})v_{it-1}$.*

The estimator, $(\widehat{\alpha}, \widehat{\rho})$, is characterized by the following system of equations defined by the moment conditions:

$$\begin{aligned} \mathbb{E}[\widehat{\xi}_{it}v_{it-1}] &= 0 \\ \mathbb{E}[\widehat{\xi}_{it}\widehat{\omega}_{it-1}] &= 0 \end{aligned} \iff \begin{aligned} (\alpha - \widehat{\alpha})\mathbb{E}[(v_{it} - \rho v_{it-1})v_{it-1}] + (\rho - \widehat{\rho})\mathbb{E}[\omega_{it-1}v_{it-1}] + (\alpha - \widehat{\alpha})(\rho - \widehat{\rho})\mathbb{E}[v_{it-1}^2] &= 0 \\ (\alpha - \widehat{\alpha})\mathbb{E}[(v_{it} - \rho v_{it-1})\omega_{it-1}] + (\rho - \widehat{\rho})\mathbb{E}[\omega_{it-1}^2] + (\alpha - \widehat{\alpha})(\rho - \widehat{\rho})\mathbb{E}[v_{it-1}\omega_{it-1}] &= 0. \end{aligned}$$

In general, the above system of equations admits two solutions. One is the true solution with $\hat{\alpha} = \alpha$ and $\hat{\rho} = \rho$, while the other solution converges to (α, ρ) as variation in the data increases. Below we formally discussed this case, but first, to understand the essence of the argument, consider the following proof sketch: when $\hat{\alpha}$ and $\hat{\rho}$ are not too far from α and ρ , respectively, the terms of the form $(\hat{\alpha} - \alpha)(\hat{\rho} - \rho)$ are of second order. In this case, the system characterizing the estimator $(\hat{\alpha}, \hat{\rho})$ reduced locally to the matrix equation $V(B - \hat{B}) = 0$, where

$$B - \hat{B} = \begin{pmatrix} \alpha - \hat{\alpha} \\ \rho - \hat{\rho} \end{pmatrix} \text{ and } V = \begin{pmatrix} \mathbb{E}[(v_{it} - \rho v_{it-1})v_{it-1}] & \mathbb{E}[\omega_{it-1}v_{it-1}] \\ \mathbb{E}[(v_{it} - \rho v_{it-1})\omega_{it-1}] & \mathbb{E}[\omega_{it-1}^2] \end{pmatrix}.$$

As long as the determinant of V is not zero, the GMM estimator is locally identified and asymptotically consistent.

Below, we show that the GMM estimator is globally identified and asymptotically consistent as long as there is enough variation in the data. The GMM estimator with AR(1) productivity (Def. 6) is characterized by the system of equations

$$\begin{cases} \mathbb{E}[\xi_{it}v_{it-1}] + (\alpha - \hat{\alpha})\mathbb{E}[(v_{it} - \rho v_{it-1})v_{it-1}] + (\rho - \hat{\rho})\mathbb{E}[\omega_{it-1}v_{it-1}] + (\alpha - \hat{\alpha})(\rho - \hat{\rho})\mathbb{E}[v_{it-1}^2] = 0 \\ \mathbb{E}[\xi_{it}\omega_{it-1}] + (\alpha - \hat{\alpha})\mathbb{E}[(v_{it} - \rho v_{it-1})\omega_{it-1}] + (\rho - \hat{\rho})\mathbb{E}[\omega_{it-1}^2] + (\alpha - \hat{\alpha})(\rho - \hat{\rho})\mathbb{E}[v_{it-1}\omega_{it-1}] = 0 \end{cases}$$

$$\iff \begin{cases} g + aX + bY + cXY = 0 \\ h + dX + eY + fXY = 0, \end{cases}$$

where $X = \alpha - \hat{\alpha}$, $Y = \rho - \hat{\rho}$, and, $a = \mathbb{E}[(v_{it} - \rho v_{it-1})v_{it-1}]$, $b = \mathbb{E}[\omega_{it-1}v_{it-1}]$, $c = \mathbb{E}[v_{it-1}^2]$, $d = \mathbb{E}[(v_{it} - \rho v_{it-1})\omega_{it-1}]$, $e = \mathbb{E}[\omega_{it-1}^2]$, $f = \mathbb{E}[v_{it-1}\omega_{it-1}]$, $g = \mathbb{E}[\xi_{it}v_{it-1}]$, and $h = \mathbb{E}[\xi_{it}\omega_{it-1}]$. Let us look at the asymptotic where $g = 0$ and $h = 0$. Assuming $c \neq 0$, we get

$$\begin{cases} aX + bY + cXY = 0 \\ dX + eY + fXY = 0 \end{cases} \iff \begin{cases} X = 0 \\ Y = 0 \end{cases} \text{ or } \begin{cases} X = -\frac{bd-ae}{cd-af} \\ Y = \frac{bd-ae}{ce-bf} \end{cases} \text{ if } cd - af \neq 0 \text{ and } ce - bf \neq 0.$$

It follows that there are two global solutions for the GMM estimator with AR(1):

$$\begin{cases} \hat{\alpha} = \alpha \\ \hat{\rho} = \rho \end{cases} \text{ or } \begin{cases} \hat{\alpha} = \alpha - \frac{bd-ae}{cd-af} = \alpha - \sqrt{\frac{\text{Var}[\omega_{it-1}]}{\text{Var}[v_{it-1}]}} \frac{\text{Corr}(\tilde{v}_{it}, v_{it-1}) - \text{Corr}(\tilde{v}_{it}, \omega_{it-1})\text{Corr}(\omega_{it-1}, v_{it-1})}{\text{Corr}(\tilde{v}_{it}, \omega_{it-1}) - \text{Corr}(\tilde{v}_{it}, v_{it-1})\text{Corr}(\omega_{it-1}, v_{it-1})} \\ \hat{\rho} = \rho + \frac{bd-ae}{ce-bf} = \rho + \sqrt{\frac{\text{Var}[\tilde{v}_{it}]}{\text{Var}[v_{it-1}]}} \frac{\text{Corr}(\tilde{v}_{it}, v_{it-1}) - \text{Corr}(\tilde{v}_{it}, \omega_{it-1})\text{Corr}(\omega_{it-1}, v_{it-1})}{1 - \text{Corr}(\omega_{it-1}, v_{it-1})^2}, \end{cases}$$

where $\tilde{v}_{it} \equiv v_{it} - \rho v_{it-1} = \frac{1}{1-\alpha}(\xi_{it} + mc_{it} - \rho mc_{it-1} + w_t - \rho w_{t-1})$.⁴

The GMM estimator admits (exactly) two possible solutions. One solution provides the true value of the parameters, while the second solution is unrelated to the true parameters. However, if $\text{Var}[v_{it-1}]$ is large compared to $\text{Var}[\omega_{it-1}]$ and $\text{Var}[\tilde{v}_{it}]$ (that is, their ratio goes to infinity while keeping fixed the correlation structure), then there is a unique solution for $\hat{\alpha}$ and $\hat{\rho}$. To conclude, if there is enough variation in the data, the GMM estimator is identified.

A.4.4. Full Proof

In this appendix, we study the production function estimator for an arbitrary number of inputs, an arbitrary functional form (Cobb-Douglas or Translog), and an AR(1) productivity process. Specifically, we assume the output of firm i at time t is such that $y_{it} = X'_{it}\beta + \omega_{it}$,

⁴Note that $\text{Corr}(\tilde{v}_{it}, \omega_{it-1}) = \text{Corr}(mc_{it} - \rho mc_{it-1} + w_t - \rho w_{t-1}, \omega_{it-1})$. Intuitively, if input price and marginal cost ($= P_{it}/\mu_{it}$) are uncorrelated with past values of productivity, this correlation will be equal to zero.

where $\beta \in \mathbb{R}^N$ is a vector of parameters to be estimated, and $X_{it} \in \mathbb{R}^N$ is a vector of inputs that can contain moments and products of several inputs. This formulation nests the Cobb-Douglas and Translog case. For example, a two-inputs v_{it}, k_{it} translog production function is modelled by $X_{it} = (v_{it}, k_{it}, v_{it}^2, k_{it}^2, v_{it}k_{it})'$ with parameters $\beta = (\beta_v, \beta_k, \beta_{vv}, \beta_{kk}, \beta_{vk})'$. We further assume that the (log) productivity ω_{it} follows an AR(1) process; that is, $\omega_{it} = \rho\omega_{it-1} + \xi_{it}$. The GMM estimator that we study here is defined as follows:

DEFINITION 7: *The GMM estimator is $\hat{\beta} \in \mathbb{R}^N$ and $\hat{\rho} \in \mathbb{R}$, so moments $\mathbb{E} [X_{it-1}\hat{\xi}_{it}]$ and $\mathbb{E} [\hat{\omega}_{it-1}\hat{\xi}_{it}]$ are equal to zero where $\hat{\omega}_{it} = y_{it} - X'_{it}\hat{\beta} = X'_{it}(\beta - \hat{\beta}) + \omega_{it}$ and $\hat{\xi}_{it} = \hat{\omega}_{it} - \hat{\rho}\hat{\omega}_{it-1} = (X_{it} - \rho X_{it-1})'(\beta - \hat{\beta}) + X'_{it-1}(\beta - \hat{\beta})(\rho - \hat{\rho}) + \omega_{it-1}(\rho - \hat{\rho}) + \xi_{it}$.*

The remainder of this appendix studies the condition under which the above estimator admits solutions. To this end, let us study the following system of equations, which defined the estimator and whose unknowns are $\hat{\beta}$ and $\hat{\rho}$:

$$\begin{cases} \mathbb{E} [X_{it-1}\hat{\xi}_{it}] = 0 \\ \mathbb{E} [\hat{\omega}_{it-1}\hat{\xi}_{it}] = \mathbb{E} [X_{it-1}\hat{\xi}_{it}]'(\beta - \hat{\beta}) + \mathbb{E} [\omega_{it-1}\hat{\xi}_{it}] = 0 \end{cases} \iff \begin{cases} \mathbb{E} [X_{it-1}\hat{\xi}_{it}] = 0 \\ \mathbb{E} [\omega_{it-1}\hat{\xi}_{it}] = 0 \end{cases} \iff$$

$$\begin{cases} \mathbb{E} [X_{it-1}\tilde{X}'_{it}](\beta - \hat{\beta}) + \mathbb{E} [X_{it-1}X'_{it-1}](\beta - \hat{\beta})(\rho - \hat{\rho}) + \mathbb{E} [X_{it-1}\omega_{it-1}](\rho - \hat{\rho}) = 0 \\ \mathbb{E} [\omega_{it-1}\tilde{X}'_{it}](\beta - \hat{\beta}) + \mathbb{E} [\omega_{it-1}X'_{it-1}](\beta - \hat{\beta})(\rho - \hat{\rho}) + \mathbb{E} [\omega_{it-1}^2](\rho - \hat{\rho}) = 0 \end{cases},$$

where we use $\mathbb{E} [X_{it-1}\xi_{it}] = 0$ and $\mathbb{E} [\omega_{it-1}\xi_{it}] = 0$, and where we denote $\tilde{X}_{it} = X_{it} - \rho X_{it-1}$. Note that the first line of the above system of equations corresponds to N equations, while the second line is just a scalar equation. We have $N + 1$ equations with unknown $(\hat{\beta}, \hat{\rho}) \in \mathbb{R}^{N+1}$. In general, this system of equations has multiple solutions, as in the case of one input.

Heuristically, when $(\hat{\beta}, \hat{\rho})$ is not too far from the true value (β, ρ) , the terms in $(\beta - \hat{\beta})(\rho - \hat{\rho})$ are of second order. Ignoring these terms leads to the following reduced system, which can be written in matrix form:

$$\begin{cases} \mathbb{E} [X_{it-1}\tilde{X}'_{it}](\beta - \hat{\beta}) + \mathbb{E} [X_{it-1}\omega_{it-1}](\rho - \hat{\rho}) = 0 \\ \mathbb{E} [\omega_{it-1}\tilde{X}'_{it}](\beta - \hat{\beta}) + \mathbb{E} [\omega_{it-1}^2](\rho - \hat{\rho}) = 0 \end{cases} \iff \begin{pmatrix} \mathbb{E} [X_{it-1}\tilde{X}'_{it}] & \mathbb{E} [X_{it-1}\omega_{it-1}] \\ \mathbb{E} [\omega_{it-1}\tilde{X}'_{it}] & \mathbb{E} [\omega_{it-1}^2] \end{pmatrix} \begin{pmatrix} \beta - \hat{\beta} \\ \rho - \hat{\rho} \end{pmatrix} = 0,$$

which admits a unique solution $(\hat{\beta}, \hat{\rho}) = (\beta, \rho)$ as long as the $(N \times N)$ matrix $\begin{pmatrix} \mathbb{E} [X_{it-1}\tilde{X}'_{it}] & \mathbb{E} [X_{it-1}\omega_{it-1}] \\ \mathbb{E} [\omega_{it-1}\tilde{X}'_{it}] & \mathbb{E} [\omega_{it-1}^2] \end{pmatrix}$, is invertible. We conclude that the GMM estimator is locally consistently identified.

Using Revenue Instead of Quantity. The estimator for revenue is as follows:

DEFINITION 8: *The GMM estimator is $\hat{\beta} \in \mathbb{R}^N$ and $\hat{\rho} \in \mathbb{R}$ such that the moments $\mathbb{E} [X_{it-1}\hat{\varsigma}_{it}]$ and $\mathbb{E} [\widehat{\text{tfpr}}_{it-1}\hat{\varsigma}_{it}]$ are equal to zero where $\widehat{\text{tfpr}}_{it} \equiv \omega_{it} + p_{it}$, $\widehat{\text{tfpr}}_{it} = r_{it} - X'_{it}\hat{\beta} = X'_{it}(\beta - \hat{\beta}) + \text{tfpr}_{it}$ and $\hat{\varsigma}_{it} = \widehat{\text{tfpr}}_{it} - \hat{\rho}\widehat{\text{tfpr}}_{it-1} = (X_{it} - \rho X_{it-1})'(\beta - \hat{\beta}) + X'_{it-1}(\beta - \hat{\beta})(\rho - \hat{\rho}) + \text{tfpr}_{it-1}(\rho - \hat{\rho}) + p_{it} - \rho p_{it-1} + \xi_{it}$.*

Using the same notation as above and $\tilde{p}_{it} = p_{it} - \rho p_{it-1}$, the system of equations that characterized the above estimator is given by

$$\begin{cases} \mathbb{E} [X_{it-1} \hat{\varsigma}_{it}] = 0 \\ \mathbb{E} [\widehat{\text{tfpr}}_{it-1} \hat{\varsigma}_{it}] = \mathbb{E} [X_{it-1} \hat{\varsigma}_{it}]' (\beta - \hat{\beta}) + \mathbb{E} [\text{tfpr}_{it-1} \hat{\varsigma}_{it}] = 0 \end{cases} \iff \begin{cases} \mathbb{E} [X_{it-1} \hat{\varsigma}_{it}] = 0 \\ \mathbb{E} [\text{tfpr}_{it-1} \hat{\varsigma}_{it}] = 0 \end{cases} \iff$$

$$\begin{cases} \mathbb{E} [X_{it-1} \tilde{X}'_{it}] (\beta - \hat{\beta}) + \mathbb{E} [X_{it-1} X'_{it-1}] (\beta - \hat{\beta})(\rho - \hat{\rho}) + \mathbb{E} [X_{it-1} \text{tfpr}_{it-1}] (\rho - \hat{\rho}) + \mathbb{E} [X_{it-1} \tilde{p}_{it}] = 0 \\ \mathbb{E} [\text{tfpr}_{it-1} \tilde{X}'_{it}] (\beta - \hat{\beta}) + \mathbb{E} [\text{tfpr}_{it-1} X'_{it-1}] (\beta - \hat{\beta})(\rho - \hat{\rho}) + \mathbb{E} [\text{tfpr}_{it-1}^2] (\rho - \hat{\rho}) + \mathbb{E} [\text{tfpr}_{it-1} \tilde{p}_{it}] = 0, \end{cases}$$

where we use the fact that prices at $t-1$ are unrelated to the innovation ξ_{it} and thus that $\mathbb{E} [\text{tfpr}_{it-1} \xi_{it}] = \mathbb{E} [\omega_{it-1} \xi_{it}] + \mathbb{E} [p_{it-1} \xi_{it}] = 0$. In general, this system of equations admits multiple solutions, as we show in the case of one input in Appendix A.4.3. For a heuristic proof, we abstract from the higher order terms in $(\beta - \hat{\beta})(\rho - \hat{\rho})$ which we consider small when $\hat{\beta}$ and $\hat{\rho}$ are not too far from their true values. In that case, the system of equations that characterized the estimator can be written in a matrix form as $W(B - \hat{B}) + R = 0$, where $B - \hat{B} = (\beta - \hat{\beta}, \rho - \hat{\rho})'$;

$$R = \begin{pmatrix} \mathbb{E} [X_{it-1} \tilde{p}_{it}] \\ \mathbb{E} [\text{tfpr}_{it-1} \tilde{p}_{it}] \end{pmatrix} \text{ and } W = \begin{pmatrix} \mathbb{E} [X_{it-1} \tilde{X}'_{it}] & \mathbb{E} [X_{it-1} \omega_{it-1}] \\ \mathbb{E} [\omega_{it-1} \tilde{X}'_{it}] & \mathbb{E} [\omega_{it-1}^2] \end{pmatrix} + \begin{pmatrix} 0 & \mathbb{E} [X_{it-1} p_{it-1}] \\ 0 & \mathbb{E} [p_{it-1} \text{tfpr}_{it-1}] \end{pmatrix},$$

which has a solution $\hat{B} = B + W^{-1}R$. As in the simple framework, the bias is due to the correlation of price (adjusted for persistence), $\tilde{p}_{it}$, with past input X_{it-1} and tfpr_{it-1} collected in the vector R .

A.5. Revenue Markup and Translog Production Function

We next compare markups from revenue and quantity production functions in a framework with a translog production function. The main intuition remains valid: the bias of the estimator on revenue data is equal to the *average* demand elasticity among firms sharing the same production function.

Assume that the production function is $y_{it} = \alpha v_{it} + \beta v_{it}^2 + \omega_{it}$, while we maintain the other assumptions of our baseline framework. Let us study the bias implied by the use of revenue data in place of quantity data. Following the same logic as above (especially as in Appendix A.4.2), the coefficients of the production function estimated on revenue are such that

$$\begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + V^{-1} \begin{pmatrix} \mathbb{E} [p_{it} v_{it-1}] \\ \mathbb{E} [p_{it} v_{it-1}^2] \end{pmatrix}, \quad \text{with} \quad V = \begin{pmatrix} \mathbb{E} [v_{it} v_{it-1}] & \mathbb{E} [v_{it}^2 v_{it-1}] \\ \mathbb{E} [v_{it} v_{it-1}^2] & \mathbb{E} [v_{it}^2 v_{it-1}^2] \end{pmatrix}.$$

As for Cobb-Douglas, the estimates are biased. The above equation is the translog equivalent of (4). Correlation of instruments with output prices causes bias.

In the case of a translog production function, the true markup is such that $\mu_{it} = (\alpha + 2\beta \log V_{it}) \frac{P_{it} Y_{it}}{W_{it} V_{it}}$, and, the revenue markup is thus $\hat{\mu}_{it}^R = \frac{\hat{\alpha} + 2\hat{\beta} \log V_{it}}{\alpha + 2\beta \log V_{it}} \mu_{it}$. As pointed out by Bond et al. (2021) and as in the Cobb-Douglas case, if we assume homogeneous inverse demand elasticities among firms in the sample (that is, for all i we have $p_{it} = -\gamma y_{it}$), the revenue markup is equal to one.⁵ However, in general, revenue markups are different from one and contain information on true markups. To see this formally, assume again that inverse demand elasticities

⁵When $p_{it} = -\gamma y_{it}$, the vector $V^{-1} \begin{pmatrix} \mathbb{E} [p_{it} v_{it-1}] \\ \mathbb{E} [p_{it} v_{it-1}^2] \end{pmatrix} = \gamma \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and the revenue markup becomes $\hat{\mu}_{it}^R = (1 - \gamma) \frac{\alpha + 2\beta \log V_{it}}{\alpha + 2\beta \log V_{it}} (1 - \gamma)^{-1} = 1$.

are heterogeneous among firms, such that for all i by $p_{it} = -d_{iit}y_{it}$ where there is at least one pair (i, j) such that $d_{iit} \neq d_{jjt}$. As above, the true markup is given by $\mu_{it} = (1 - d_{iit})^{-1}$. In this heterogeneous inverse demand elasticity case, we have

$$\begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = (I - \mathbb{E}[X_{it-1}X'_{it}])^{-1} \mathbb{E}[d_{iit}X_{it-1}X'_{it}] \begin{pmatrix} \alpha \\ \beta \end{pmatrix},$$

where X_{it} is vector $(v_{it}, v_{it-1}^2)'$, I is the identity matrix. Revenue markups satisfy

$$\hat{\mu}_{it}^R = \left[1 - (\alpha + 2\beta \log V_{it})^{-1} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}' \left(\mathbb{E}[d_{iit}X_{it}X'_{it-1}] \mathbb{E}[X_{it}X'_{it-1}]^{-1} \right) \begin{pmatrix} 1 \\ 2 \log V_{it} \end{pmatrix} \right] (1 - d_{iit})^{-1}. \quad (1)$$

This markup is in general different from one for at least some firms. To see that clearly, let us further assume that the inverse demand elasticities are independent of the variable input usage and its square, such that, for any $n, m \in \mathbb{N}$, $\mathbb{E}[d_{iit}v_{it}^n v_{it-1}^m] = \mathbb{E}[d_{iit}] \mathbb{E}[v_{it}^n v_{it-1}^m]$. With these assumptions in place, one can show that $\hat{\alpha} = \alpha(1 - \mathbb{E}[d_{iit}])$ and $\hat{\beta} = \beta(1 - \mathbb{E}[d_{iit}])$. The revenue markup is equal to $\hat{\mu}_{it}^R = (1 - \mathbb{E}[d_{iit}])^{-1}(1 - d_{iit})^{-1}$, which is different from one since there exists a pair (i, j) such that $d_{iit} \neq d_{jjt}$. As for the Cobb-Douglas case, the bias is determined by an average of the inverse demand elasticities.

In the translog case, the average revenue markup is $\mathbb{E}[\log \hat{\mu}_{it}^R] = \mathbb{E}[\log(\mu_{it})] + \mathbb{E}\left[\log \frac{\hat{\alpha} + 2\hat{\beta} \log V_{it}}{\alpha + 2\beta \log V_{it}}\right]$. Let us assume that the inverse demand elasticities are heterogeneous across firms in the sample. From equation (1), we can see that the average of the log revenue markup is equal to zero up to a Jensen-like inequality:

$$\begin{aligned} \mathbb{E}[\log \hat{\mu}_{it}^R] &= -\mathbb{E}[\log(1 - d_{iit})] + \dots \\ &\dots \mathbb{E}\left[\log \left(1 - (\alpha + 2\beta \log V_{it})^{-1} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}' \left(\mathbb{E}[d_{iit}X_{it}X'_{it-1}] \mathbb{E}[X_{it}X'_{it-1}]^{-1} \right) \begin{pmatrix} 1 \\ 2 \log V_{it} \end{pmatrix} \right) \right]. \end{aligned}$$

When the inverse demand elasticities are homogeneous, $\forall i, d_{iit} = \gamma$, then the average log revenue markup is exactly zero. In general, the relationship between the average revenue and true markup now depends on the distribution of the variable input $\log V_{it}$ and the extent of the bias in the production function estimation. Importantly, the variance of the revenue markup is different from the variance of the true markup and also depends on the distribution of inputs and the covariance of input and the true markup. Finally, the correlation between the revenue and the true markup is no longer equal to one. To gauge the information content of the revenue markup under translog, we rely on the simulations.

APPENDIX B: DERIVATION AND PARAMETRIZATION OF THE SIMULATED MODEL

B.1. Model and Parametrization

We analyze a single sector, defined as a collection of firms that have the same structural production function parameters and that face the same input prices.

Demand. We choose a market structure where firms have heterogeneous markups that are determined by a combination of structural parameters and their market share. Following [Atkeson and Burstein \(2008\)](#), we implement this by assuming that firms compete in a double-nested CES demand system. The sector consists of a continuum of markets, where a market is defined as a finite number of firms that compete oligopolistically with one another. In this setup, the

demand faced by market h , Y_{ht} , satisfies at sector price index P_{ht} :

$$P_{ht} = Y_{ht}^{-\frac{1}{\sigma}} D_t^{\frac{1}{\sigma}} \quad \text{with} \quad Y_{ht} = \left[\sum_{i=1}^{N_h} Y_{iht}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}, \quad (2)$$

where σ denotes the elasticity of substitution across markets, D_t is exogenous aggregate demand, and market-level output Y_{ht} is the aggregate of firm-level output across the N_h firms that operate in h . Y_{iht} denotes the output of firm i and ε is the elasticity of substitution across goods within a market. Following [Atkeson and Burstein \(2008\)](#), we assume $\varepsilon > \sigma$, reflecting that it is easier to substitute goods across firms than across markets. The inverse demand function for firm i is

$$P_{iht} = \left(\frac{Y_{iht}}{Y_{ht}} \right)^{-\frac{1}{\varepsilon}} P_{ht}, \quad (3)$$

where P_{iht} is the price of firm i and P_{ht} satisfies the market-level inverse demand (equation 2). Under Cournot competition, firm i in market h maximizes profit by choosing its quantity taking other firms' quantity as given – subject to the inverse demand given by the above equation (3). The quantity-setting firm internalizes that Y_{ht} increases and P_{ht} decreases when it raises its own quantities according to equation (2). The resultant profit-maximizing markup reads as

$$\mu_{iht} = \frac{\varepsilon}{\varepsilon-1} \left(1 - \frac{\frac{\varepsilon}{\sigma} - 1}{\varepsilon - 1} s_{iht} \right)^{-1} \quad \text{with} \quad s_{iht} = \frac{P_{iht} Y_{iht}}{P_{ht} Y_{ht}}, \quad (4)$$

where s_{iht} is the market share in terms of revenue.⁶ Markups range from $\varepsilon/(\varepsilon-1)$ for a firm whose market share approaches zero, to $\sigma/(\sigma-1)$ for a monopolist.

Technology. Firms produce using a variable input V_{iht} and a fixed input K_{iht} , with log-inputs respectively denoted by v_{iht} and k_{iht} . The production function for log output y_{iht} is translog:

$$y_{iht} = \omega_{it} + \gamma \alpha v_{iht} + \gamma(1-\alpha)k_{iht} + \gamma \frac{\alpha(1-\alpha)}{2} \frac{\phi-1}{\phi} (v_{iht}^2 + k_{iht}^2 - 2k_{iht}v_{iht}), \quad (5)$$

where ω_{it} is the log of (Hicks-neutral) total factor productivity, γ measures the degree of returns to scale, α determines the weight of the variable input in the production function, while ϕ approximates the elasticity of substitution between the flexible and the fixed input. When $\phi = 1$, this production function nests the Cobb-Douglas specification. Our log production function (5) is motivated by an approximation around $\phi = 1$ of the constant elasticity of substitution production function $Y_{iht} = e^{\omega_{iht}} (\alpha V_{iht}^{\frac{\phi-1}{\phi}} + (1-\alpha)K_{iht}^{\frac{\phi-1}{\phi}})^{\frac{\phi}{\phi-1}\gamma}$ (see Appendix B.2).

Variable input demand. We next derive the demand for the variable input for the translog production function. The firms' cost minimization problem involves minimizing costs $W_t V_{iht}$ subject to the production function (5). Note that the output elasticity of the variable input is $\frac{\partial y_{iht}}{\partial v_{iht}} = \gamma \alpha \left(1 + [1-\alpha] \frac{\phi-1}{\phi} [v_{iht} - k_{iht}] \right)$, such that the first-order condition of the cost minimization problem is: $W_t = MC_{iht} \frac{Y_{iht}}{V_{iht}} \gamma \alpha \left(1 + [1-\alpha] \frac{\phi-1}{\phi} \ln \left[\frac{V_{iht}}{K_{iht}} \right] \right)$, where MC_{iht} is the

⁶The Bertrand markup is similar (e.g. [Atkeson and Burstein 2008](#), [Grassi 2017](#), [Burstein et al. 2020](#)).

TABLE B.I
PARAMETER CALIBRATION FOR SIMULATION

Parameters	Value	Description
α	0.4	Share of variable input
γ	0.8	Returns to scale
ϕ	1.1	Elasticity of substitution
σ	1.1	Demand elasticity across markets
ε	10	Demand elasticity across firms in a market
N, N_h	180, 8	Number of markets and firms per market
ρ^w, σ^w	0.87, 0.06	AR(1) persistence and std. dev. of W_t
ρ^D, σ^D	0.78, 0.19	AR(1) persistence and std. dev. of $P_t^{-\sigma} Y_t$
$\rho^\omega, \sigma^\omega$	0.60, 0.20	AR(1) persistence and std. dev. of firm-level ω_{it}
ρ^k, σ^k	0.66, 0.66	AR(1) persistence and std. dev. of firm-level k_{it}
$\tilde{\sigma}^\eta$	0.095	std. dev. meas. error on output

marginal cost which can be rewritten as

$$V_{iht} = \left(\frac{MC_{iht}}{W_t} \right) \gamma \alpha \left(1 + [1 - \alpha] \frac{\phi - 1}{\phi} \ln \left[\frac{V_{iht}}{K_{iht}} \right] \right) Y_{iht}, \quad (6)$$

Marginal costs. As firms face an exogenous sequence of the fixed input K_{iht} , marginal costs can be derived from the production function (5) and optimal demand for the variable input (6). Inserting the latter into the former, we get $y_{iht} = \omega_{iht} + \gamma \alpha \ln \left[\left(\frac{MC_{iht}}{W_t} \right) \gamma \alpha \left(1 - [1 - \alpha] \frac{\phi - 1}{\phi} \ln \left[\frac{K_{iht}}{V_{iht}} \right] \right) \right]$.

Isolating log marginal costs on the left-hand side, we can express the log marginal costs $mc_{iht} \equiv \ln MC_{iht}$ as

$$mc_{iht} = \ln \left[\frac{W_t}{\gamma} Y_{iht}^{\frac{1-\alpha\gamma}{\alpha}} \Omega_{iht}^{-\frac{1}{\alpha\gamma}} K_{iht}^{\frac{\alpha-1}{\alpha}} \right] - \ln \left(1 + [1 - \alpha] \frac{\phi - 1}{\phi} \ln \left[\frac{V_{iht}}{K_{iht}} \right] \right) + \frac{1 - \alpha}{2} \frac{\phi - 1}{\phi} \left(\ln \left[\frac{V_{iht}}{K_{iht}} \right] \right)^2. \quad (7)$$

Equilibrium. We consider an equilibrium given an exogenous sequence for variable input prices W_t , aggregate demand D_t , productivities ω_{iht} and fixed factors k_{iht} . The equilibrium is defined as a sequence of markups μ_{iht} , prices P_{iht} , output Y_{iht} , log marginal costs mc_{iht} , market shares s_{iht} , log variable inputs v_{iht} , and market-level output Y_{ht} and price P_{ht} such that price is equal to markup times marginal cost, the demand is satisfied (equations 2 and 3), quantities are set to maximize profit (equations 4), and the variable input is chosen to minimize cost (equations 6 and 7 in Appendix B.2).

Calibration. We perform 200 Monte Carlo simulations. In each simulation, we model the behavior of 1600 firms, which is the average number of firms in two-sector industries in the EAP data. We divide these firms into 180 markets, the level at which firms compete, and simulate the economy for 40 periods.

There are 13 parameters, each of which we calibrate externally. The parameters are summarized in Table B.I. In calibrating the model, we are constrained by the fact that the true values of many parameters (such as those of the production function and the productivity process) are in fact the object of interest in our empirical analysis. Our approach is therefore to assume reasonable values in line with the literature as an example of a possible quantification.

There are two sources of firm heterogeneity: the firm's log-endowment of the fixed input k_{iht} and the firm's log-total factor productivity ω_{it} . Both evolve exogenously through log-linear AR(1) processes with persistence ρ_k and ρ_ω , respectively, and are subject to innovations

$\xi_k \sim N(0, \sigma_k)$ and $\xi_\omega \sim N(0, \sigma_\omega)$. Both sources of firm heterogeneity are similar in that firms with either higher productivities or higher exogenous fixed input have, *ceteris paribus*, greater output. They are different in that the fixed input is observable, while productivity is not. To calibrate the persistence and volatility of the fixed factor, we run autoregressive regressions on log capital in the data. We find a persistence parameter ρ^k of 0.66 and a volatility of shocks σ_k of 0.66. We set ρ^ω to 0.6 and set volatility σ^ω to 0.2, in line with [Decker et al. \(2020\)](#) and [Carvalho and Grassi \(2019\)](#).

There are two aggregate shocks: aggregate demand D_t and the variable input price W_t . We assume both series follow a log-linear AR(1) process with persistence ρ_D and ρ_W , respectively, and shocks $\xi_D \sim N(0, \sigma_D)$ and $\xi_W \sim N(0, \sigma_W)$. To calibrate the process for the variable inputs price, we estimate an AR(1) process for the price index of intermediate inputs from sector-level manufacturing data in EU-KLEMS. We run simple AR(1) regressions for the log of the index, and find an autoregressive coefficient ρ_W of 0.87 at the two-digit sector level when controlling for industry- and year-fixed effects. Residuals have a standard deviation σ_W of 0.06. For aggregate demand D_t we estimate a similar autoregressive process, using the detrended sector-level nominal value added as the dependent variable.⁷ We find a high degree of persistence in aggregate demand, with a ρ^D of 0.78, while the residuals have a standard deviation of 0.19.

When calibrating the production function, we think of purchased materials as v_{iht} and a composite of all other factors as k_{iht} . We calibrate α to 0.4 to match the average ratio of material purchases over revenue in EAP-FARE, which is 0.38. We calibrate returns-to-scale parameter γ to 0.8 in order to have modest decreasing returns to scale, in line with the estimate by [Basu and Fernald \(1997\)](#). We assume an elasticity of substitution ϕ of 1.1, as purchased materials include intermediate inputs from other firms, which can substitute for in-house production.

We introduce measurement error in observed quantity $\tilde{y}_{iht}$, denoted by η_{iht} , after computing the equilibrium. We assume that $\eta_{iht} \sim N(0, \sigma_y \tilde{\sigma}_\eta)$, where σ_y is the standard deviation of true output across all firm-years in the sector, and $\tilde{\sigma}_\eta$ is a scalar that determines the magnitude of measurement error relative to the standard deviation of true output. To calibrate the degree of measurement error in output, $\tilde{\sigma}_\eta$, we perform a regression using the EAP micro data that is identical to the first stage of the baseline markup estimation. The dependent variable is quantity, and the explanatory variables are prices, market share, time-fixed effects and a third-degree polynomial in the firms' inputs. The R^2 is 0.905, which implies that the residual – which captures measurement error – explains 9.5% of the variation in output. We calibrate $\tilde{\sigma}_\eta$ so that measurement error explains 9.5% of the variance of output in the simulations as well.

B.2. Translog Approximation

This appendix derives the translog production function (equation 5) as an approximation of a CES production function around $\phi = 1$ with homogeneity of degree γ that is $Y_{iht} = \Omega_{iht}(\alpha[V_{iht}]^{\frac{\phi-1}{\phi}} + (1-\alpha)[K_{iht}]^{\frac{\phi-1}{\phi}})^{\frac{\phi}{\phi-1}\gamma}$. To see this, rearrange terms

$$\begin{aligned} \ln y_{iht} &= \omega_{iht} + \frac{\phi}{\phi-1} \gamma \ln \left[\alpha[V_{iht}]^{\frac{\phi-1}{\phi}} + (1-\alpha)[K_{iht}]^{\frac{\phi-1}{\phi}} \right] = \omega_{iht} + \frac{\phi}{\phi-1} \gamma \ln \left[\alpha[V_{iht}]^{\frac{\phi-1}{\phi}} \left(1 + \frac{(1-\alpha)}{\alpha} \left[\frac{K_{iht}}{V_{iht}} \right]^{\frac{\phi-1}{\phi}} \right) \right] \\ &= \omega_{iht} + \frac{\phi}{\phi-1} \gamma \ln \left[\alpha[V_{iht}]^{\frac{\phi-1}{\phi}} \right] + \frac{\phi}{\phi-1} \gamma \ln \left[1 + \frac{1-\alpha}{\alpha} \left(\frac{K_{iht}}{V_{iht}} \right)^{\frac{\phi-1}{\phi}} \right] = \omega_{iht} + \gamma v_{iht} + \frac{\phi}{\phi-1} \gamma \ln \left[\alpha + (1-\alpha) \left(\frac{K_{iht}}{V_{iht}} \right)^{\frac{\phi-1}{\phi}} \right], \end{aligned}$$

⁷We detrend D_t using nominal GDP to account both for increases in prices and real output to obtain a stationary nominal series. Results are similar when detrending with the GDP deflator.

where we move the α back into the log term for the last equality. Rewriting the final term yields

$\frac{\phi}{\phi-1}\gamma \ln \left[1 + (1-\alpha) \left(\left(\frac{K_{iht}}{V_{iht}} \right)^{\frac{\phi-1}{\phi}} - 1 \right) \right]$; let us then define $f(x) = \frac{\gamma}{x} \ln [1 + (1-\alpha)(B^x - 1)]$, where $B = K_{iht}/V_{iht}$ and $x = (\phi-1)/\phi$, such that our approximation is around $x \rightarrow 0$. Taking a second-order approximation yields

$$\begin{aligned} f(x) &= \frac{\gamma}{x} \ln [1 + (1-\alpha)((\exp(x \ln B) - 1))] \approx \frac{\gamma}{x} \ln \left[1 + (1-\alpha) \left(x \ln B - \frac{x^2 [\ln B]^2}{2} \right) \right] \\ &\approx \frac{\gamma}{x} \left[(1-\alpha) \left(x \ln B - \frac{x^2 [\ln B]^2}{2} \right) - \frac{(1-\alpha)^2}{2} \left(x \ln B - \frac{x^2 [\ln B]^2}{2} \right)^2 \right] \\ &\approx \frac{\gamma}{x} \left[(1-\alpha)x \ln B + \alpha \frac{1-\alpha}{2} x^2 [\ln B]^2 \right], \end{aligned}$$

where we remove higher-order terms given that we are approximating the function up to a second order. Hence, the first-order approximation of the generalized CES production function reads $y_{iht} = \omega_{iht} + \gamma \ln V_{iht} + \gamma(1-\alpha) \ln \left(\frac{K_{iht}}{V_{iht}} \right) + \gamma\alpha \frac{1-\alpha}{2} \frac{\phi-1}{\phi} \left[\ln \left(\frac{K_{iht}}{V_{iht}} \right) \right]^2$. After rearranging terms and denoting by small cap letters the log of a variable, $x \equiv \ln X$, we have the translog production function (5) with homogeneity of degree γ : $y_{iht} = \omega_{iht} + \gamma\alpha v_{iht} + \gamma(1-\alpha)k_{iht} + \gamma\alpha \frac{1-\alpha}{2} \frac{\phi-1}{\phi} (v_{iht}^2 + k_{iht}^2 - 2k_{iht}v_{iht})$.

APPENDIX C: IMPLEMENTATION OF THE ESTIMATION

This appendix describes how we implement the production function estimation. Let us assume that the observed output of firm i at time t is such that $\tilde{y}_{it} = X'_{it}\beta + \omega_{it} + \eta_{it}$, where $\beta \in \mathbb{R}^N$ is a vector of parameters to be estimated, and $X_{it} \in \mathbb{R}^N$ is a vector of inputs that can contain monomes and products of several inputs. This formulation nests the Cobb-Douglas and Translog case. We assume further that η_{it} is a measurement error shock, such that actual output is $y_{it} = \tilde{y}_{it} - \eta_{it} = X'_{it}\beta + \omega_{it}$. Productivity, ω_{it} , follows a Markov process. We assume that we have access to a sample of observed output $\tilde{y}_{it}$ and input usage X_{it} . Here we assume that we additionally observe price p_{it} and controls for markups s_{it} .

The first stage consists of purging the observed output from measurement errors. As explained in Section 2.3, we do so by running a regression of measure output $\tilde{y}_{it}$ on time-fixed effects, a polynomial of inputs usage of some order (second or third in practice), price p_{it} , and the additional controls s_{it} .⁸ We then compute an estimate of the measurement errors $\hat{\eta}_{it}$ as the difference of observed output $\tilde{y}_{it}$ and the fitted value $\hat{y}_{it}$, which we interpret as true output.

The estimator of the parameters $\hat{\beta}$ is delivered by a numerical algorithm that makes moments equal to zero. These moments are computed as follows. For a given guess of parameters $\hat{\beta}$, we compute $\hat{\omega}_{it} = y_{it} - \hat{\beta}'X_{it}$. We then estimate the Markov process: in the case of AR(1), we obtain $\hat{\rho}$ by an OLS regression. The estimates of the innovation of the AR(1) process are given by $\hat{\xi}_{it} = \hat{\omega}_{it} - \hat{\rho}\hat{\omega}_{it-1}$. The set of instruments Z_{it} is chosen from the vector of input usage X_{it} or its lag for dynamic and static input, respectively (see Appendix A.4.2 for an example). Finally, we compute the moments as $\sum_{i,t} \hat{\xi}_{it} Z_{it}$.

⁸When price and quantity are not observed, revenue is used in place of observed output; we do not include the extras controls p_{it} and s_{it} in the first-stage regression.

APPENDIX D: CONVERGENCE

We assess the convergence properties of the estimators. In particular, we examine the speed at which our markup estimates converge to their true values as sample size increases. In the baseline calibration, we simulate 1600 firms per sector, matching the average sector size in France. However, many sectors have fewer firms in practice. To determine if markups can be reliably estimated in smaller samples, we repeat our estimations for samples ranging from 150 to 1600 firms per year, increasing in increments of 50. Results indicate that precision in estimating the average log markup improves significantly up to approximately 500-600 firms, beyond which additional increases in sample size yield limited improvements. Since most sectors in administrative datasets surpass this threshold, accurately estimating markups appears feasible with administrative datasets that are typically available.

APPENDIX E: ADDITIONAL TABLES AND FIGURES

TABLE E.I
ESTIMATED MATERIAL-OUTPUT ELASTICITY FOR VARIOUS SPECIFICATIONS BY SECTOR

Industry	NACE	Quantity		Revenue	
		Baseline	BB-Q	ACF	BB-R
All (average)	-	0.54 (0.29)	0.38 (0.30)	0.40 (0.15)	0.39 (0.14)
Manufacturing of textiles	13	0.45 (0.23)	0.22 (0.14)	0.41 (0.16)	0.44 (0.13)
Manufacturing of wearing apparel	14	0.46 (0.25)	0.36 (0.16)	0.27 (0.20)	0.28 (0.14)
Manufacturing of leather and related products	15	0.38 (0.22)	0.34 (0.20)	0.30 (0.19)	0.26 (0.14)
Manufacturing of wood and products of wood	16	0.58 (0.15)	0.64 (0.24)	0.47 (0.14)	0.49 (0.14)
Manufacturing of paper and paper products	17	0.52 (0.19)	0.35 (0.09)	0.46 (0.12)	0.45 (0.09)
Manufacturing of printing and reproduction	18	0.34 (0.20)	0.19 (0.12)	0.41 (0.13)	0.41 (0.13)
Manufacturing of chemicals and chemical products	20	0.68 (0.40)	0.38 (0.24)	0.48 (0.16)	0.40 (0.12)
Manufacturing of rubber and plastic products	22	0.60 (0.18)	0.41 (0.11)	0.45 (0.13)	0.49 (0.15)
Manufacturing of other non-metallic mineral products	23	0.53 (0.13)	0.50 (0.25)	0.39 (0.15)	0.34 (0.10)
Manufacturing of basic metals	24	0.62 (0.19)	0.72 (0.15)	0.43 (0.19)	0.43 (0.17)
Manufacturing of fabricated metal products	25	0.42 (0.19)	0.28 (0.41)	0.37 (0.15)	0.31 (0.11)
Manufacturing of computer-, electronic products	26	0.71 (0.31)	0.26 (0.31)	0.39 (0.13)	0.34 (0.11)
Manufacturing of electrical equipment	27	0.62 (0.22)	0.42 (0.17)	0.46 (0.14)	0.48 (0.14)
Manufacturing of machinery and equipment	28	0.37 (0.21)	0.53 (0.38)	0.43 (0.13)	0.36 (0.09)
Manufacturing of motor vehicles	29	0.81 (0.32)	0.65 (0.22)	0.52 (0.17)	0.53 (0.14)
Manufacturing of furniture	31	1.15 (0.15)	0.31 (0.34)	0.39 (0.10)	0.38 (0.06)
Manufacturing of other	32	0.43 (0.28)	0.24 (0.14)	0.30 (0.13)	0.30 (0.12)
Repair and installation of machinery and equipment	33	0.31 (0.13)	0.14 (0.14)	0.31 (0.11)	0.31 (0.11)

NOTE: Estimated elasticities of materials on output from the estimation of translog production functions. The headers, "Baseline", "BB-Q", "ACF" and "BB-R", refers to different specifications. "Baseline": IV-GMM on observed quantity. "BB-Q" and "BB-R": dynamic panel estimators. "ACF": IV-GMM on revenue. See Section 5.1 for details. Translog specifications have heterogeneous elasticities within industries, with standard deviations presented in brackets. Industry codes refer to two-digit NACE codes. Industry names are provided in Table G.II.

TABLE E.II
SUMMARY STATISTICS - WITHIN-SECTOR MOMENTS OF THE LOG MARKUP ESTIMATES

	Mean	St. Dev.	Median*	25th Pct.*	75th Pct.*
Quantity	-0.003 (0.007)	0.536 (0.006)	0.040 (0.009)	-0.312 (0.011)	0.354 (0.009)
Revenue	0.072 (0.003)	0.210 (0.002)	-0.026 (0.004)	-0.150 (0.003)	0.124 (0.004)

NOTE: Moments are calculated at the sector level and then averaged across the sectors in the data. Single-stage markup estimates. Parentheses present bootstrapped standard errors of the moment. 147,704 obs. *: Medians and percentiles are expressed in deviation of sector averages.

TABLE E.III
CORRELATION BETWEEN QUANTITY AND REVENUE-BASED MARKUPS LOG MARKUPS

	Summary of Within-Sector Correlations					Pooled
	Average	St. Dev.	Median	25th Pct.	75th Pct.	Overall
Pearson Correlation	0.39	(0.35)	0.45	0.19	0.62	0.12
Pearson Corr. First Diff.	0.55	(0.29)	0.58	0.36	0.78	0.41
Rank Correlation	0.40	(0.35)	0.50	0.25	0.65	0.12
Rank Corr. First Diff.	0.60	(0.25)	0.60	0.40	0.80	0.54

NOTE: The table presents correlations between single-stage revenue- and quantity-based markup estimates. Correlations are calculated at the two-digit sector in the left-hand panel, and for all observations pooled in the right-hand panel. The summary statistics are averages of the correlations across sectors. 147,704 observations.

APPENDIX F: CORRELATION PRICES AND INSTRUMENTS

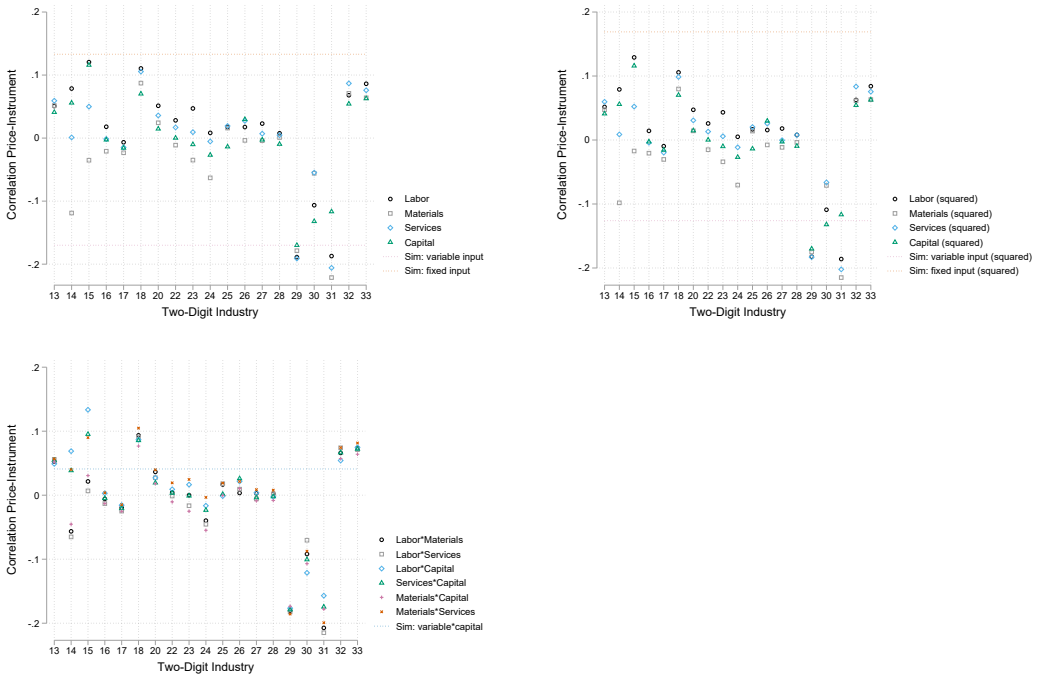
In this appendix we present the correlation between prices and the production function instruments. Section 2.2.4 shows that revenue-based markup estimates for the translog production function will be close to true markups if the correlations between prices and the instruments are low. That result is derived for i.i.d. productivity. If productivity is linear first-order autoregressive, correlation between the instruments and $\tilde{p}_{it} \equiv p_{it} - \rho p_{it-1}$ (where ρ is the persistence of productivity) determines the correlation between true markups and revenue-based markups. Hence, all correlations here are between $\tilde{p}_{it}$ and the production function estimation instruments.

TABLE F.I
CORRELATION OF INSTRUMENTS AND PRICE IN THE SIMULATIONS

Instruments	v_{iht-1}	k_{iht}	v_{iht-1}^2	k_{iht}^2	$v_{iht-1}k_{iht}$
Instrument's correlation with $\tilde{p}_{it}$	.134 (.004)	-.177 (.004)	-.127 (.004)	.169 (.004)	.040 (.003)

NOTE: $\tilde{p}_{it} \equiv p_{it} - \rho p_{it-1}$ where ρ is the persistence of productivity. The inputs are denoted by v for variable, k for fixed. The table presents average correlations across the Monte Carlo, simulations while parentheses give the standard deviation.

FIGURE F.1.—Correlation between Prices and Instruments



NOTE: The figure plots the correlation between $\tilde{p}_{it} \equiv p_{it} - \rho p_{it-1}$ where ρ is the persistence of productivity, and instruments in the production function estimation. The instruments are l_{it} , k_{it} , m_{it-1} , o_{it-1} , l_{it}^2 , k_{it}^2 , m_{it-1}^2 , o_{it-1}^2 respectively for the inputs l_{it} , k_{it} , m_{it} , o_{it} , l_{it}^2 , k_{it}^2 , m_{it}^2 , o_{it}^2 , where inputs are denoted by l for labor, k for capital, o for service, and m for material. The horizontal axes give the two-digit industry for which correlations are calculated. Legends give the input for which the instrument's correlation is plotted.

Table F.I presents the correlations for the baseline simulation. The instruments for the variable input are its lags, as the identification assumption is that innovations in productivity are orthogonal to the lagged variable input. The instruments for the fixed factor are just its contemporaneous value, as the fixed factor is determined exogenously and thus uncorrelated with productivity innovations to begin with. We calculate the correlations for each of the Monte Carlo repetitions and present the average correlations in the table. The table shows that the correlations between prices and the instruments are generally close to zero, ranging from 0.041 for the interaction term to -0.177 for the square of the fixed factor.

The correlations in the simulations are similar to the correlations in the data. In the data there are more inputs (and thus instruments); since we estimate the production function separately for every sector, the correlations are plotted in Figure F.1. Horizontal axes give the two-digit

industry codes. From left to right, the figures contain correlations between price innovations and instruments for linear inputs; for quadratic inputs; and for interacted inputs. Each dot presents a sector level correlation. Horizontal lines, for reference, give the correlations in the simulations.

The figure shows that the empirical correlations between price innovations and instruments are either similar to or closer to zero than the simulations. As a result, in the absence of empirical complications such as misspecification of the production function or heterogeneous production functions within sectors, the correlations between revenue-based markup estimates and quantity-based markup estimates in the data should be similar to the simulations.

APPENDIX G: DATA CONSTRUCTION

The dataset is constructed from two sources. The first is a panel of the universe of French firms, which we obtain from [Burstein et al. \(2020\)](#).

The panel is based on FARE, which is an administrative dataset based on data from the tax office DGFIP. It contains detailed balance-sheet- and income statement information. We also drop firms with fewer than two employees, as the number of single-employee firms grew rapidly over our sample due to a regulatory change. The FARE data contains all of the variables for the production function estimation with the exception of quantities and prices.

For revenue, we use total sales (including exports). This is REDI_R310 in FARE. The wage bill is the sum of wage payments and social security contributions (REDI_R216 and REDI_R217). Materials, the variable input in our empirical section, is the sum of intermediate inputs and raw materials, adjusted for inventories (REDI_R210, REDI_R212, REDI_R211, and REDI_213). Services include other purchases, such as marketing and insurance (REDI_214). The capital stock is measured as fixed tangible assets (IMMO_CORP in FARE). Market share is defined as a firm's total share in sales in its five-digit NACE industry, which we calculate before outlier treatments are applied. We drop firms with missing, zero or negative revenue, material purchases, service purchases, wage bills, capital or prices and winsorize variables within two-digit industries, and deflate nominal variables using EU-KLEMS deflators. As in the analytical model, we assume firms within sectors face equal factor prices.

We merge the panel with the administrative survey *Enquête Annuelle de Production* (EAP). EAP is a product-level survey by the statistical office (INSEE) and is the French counterpart of the European PRODCOM survey. It details revenue and quantity sold by 10-digit product code for the universe of manufacturing firms with at least 20 employees or 5 million euros in revenue. The sample is from 2009 to 2019. We merge EAP to FARE using the *siren* code, which is a common firm identifier.

The baseline sample is the intersection of the EAP and the FARE dataset. We include all firm-products unless the firm only acted as an outsourcer or designer, and drop around one-third of firm-products without quantity data. We define a product as a combination of a 10-digit product code and a unit of account, because products reported in different units may not be comparable. For each firm-product we calculate the ratio of revenue over the quantity of the product sold. We then standardize this unit value by dividing it by the revenue-weighted average price of the 10-digit product across the sample. As some firms produce multiple products, we define a firm's price as the sales-weighted average of standardized prices across the goods that it produces. Quantity is the ratio of revenue over this price.⁹ Summary statistics are given in Tables [G.I](#) and [G.II](#).

⁹As a robustness check we standardize prices using the revenue-weighted average price at the 8-digit sector level. These firm-level prices have a 0.89 correlation with our baseline prices.

TABLE G.I
SUMMARY STATISTICS

<i>Variable</i>	Mean	St. Dev.	Median	10th Pct.	90th Pct.	Observations
<i>FARE</i>						
Revenue	15,117	53,088	3,045	544	30,619	175,538
Wage Bill	3,008	10,268	830	194	6,505	175,538
Capital	7,245	27,096	869	114	13,416	175,538
Purchased Materials	6,622	22,433	1,017	116	13,615	175,538
Purchased Services	3,770	19,599	755	120	7,388	175,538
<i>EAP</i>						
Quantity	13,006	47,757	1,891	236	27,458	175,538
Standardized Price	5.66	42.21	1.23	0.77	6.25	175,538

NOTE: Nominal values are in thousands of 2010 euros, deflated using EU-KLEMS deflators. Revenue is deflated with the gross output deflator; purchased inputs are deflated using the intermediate input deflator. Wages and capital stock are deflated using the GDP deflator. Capital is measured as fixed tangible assets. ¹⁰ The data contains 26,143 unique firms across 206 (19) sectors at the five (two) digit level.

TABLE G.II
SECTORS (TWO-DIGIT) IN THE EAP-FARE DATASET

Manufacturing of ...	NACE code	Observations
... textiles	13	6,716
... wearing apparel	14	5,200
... leather and related products	15	2,256
... wood and products of wood and cork, except furniture	16	9,599
... paper and paper products	17	6,511
... printing and reproduction of recorded media	18	8,589
... chemicals and chemical products	20	8,498
... rubber and plastic products	22	17,939
... other non-metallic mineral products	23	13,850
... basic metals	24	4,471
... fabricated metal products, except machinery and equipment	25	26,693
... computer, electronic and optical products	26	6,401
... electrical equipment	27	7,575
... machinery and equipment n.e.c.	28	16,738
... motor vehicles, trailers and semi-trailers	29	5,493
... other transport equipment	30	889
... furniture	31	10,844
... other	32	5,094
Repair and installation of machinery and equipment	33	12,182

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